

JAVA WITH DSA-D3

MATRIX ADDITION:

```
import java.util.*;

class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int a[][] = new int[3][3];
        int b[][] = new int[3][3];
        int c[][] = new int[3][3];

        System.out.println("Enter the first matrix:");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                a[i][j] = sc.nextInt();
            }
        }

        System.out.println("Enter the second matrix:");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                b[i][j] = sc.nextInt();
            }
        }

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                c[i][j] = a[i][j] + b[i][j];
            }
        }

        System.out.println("Result");
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(c[i][j] + " ");
            }

            System.out.println();
        }
    }
}
```

```
}
```

OUTPUT:

Enter the first matrix:

1 2 3

4 5 6

7 8 9

Enter the second matrix:

9 8 7

6 5 4

3 2 1

Result

10 10 10

10 10 10

10 10 10

TRANSPOSE:

```
import java.util.*;
```

```
class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int a[][] = new int[3][3];
```

```
        int t[][] = new int[3][3];
```

```
        System.out.println("Enter the matrix:");
```

```
        for (int i = 0; i < 3; i++) {
```

```
            for (int j = 0; j < 3; j++) {
```

```
                a[i][j] = sc.nextInt();
```

```
            }
```

```
        }
```

```
        System.out.println("Transpose:");
```

```
        for (int i = 0; i < 3; i++) {
```

```
            for (int j = 0; j < 3; j++) {
```

```
                t[i][j] = a[j][i];
```

```
            }
```

```
        }
```

```
        System.out.println("Result");
```

```

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(t[i][j] + " ");
            }
            System.out.println();
        }
    }
}

```

OUTPUT:

Enter the matrix:

1 2 3

4 5 6

7 8 9

Transpose:

Result

1 4 7

2 5 8

3 6 9

SPIRAL ORDER:

```
import java.util.*;
```

```
class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int a[][] = new int[3][3];
```

```
        System.out.println("Enter matrix elements:");
```

```
        for (int i = 0; i < 3; i++) {
```

```
            for (int j = 0; j < 3; j++) {
```

```
                a[i][j] = sc.nextInt();
```

```
            }
```

```
        }
```

```
        int top = 0, bottom = 2;
```

```
        int left = 0, right = 2;
```

```
        System.out.println("Spiral Order:");
```

```
        while (top <= bottom && left <= right) {
```

```

    for (int i = left; i <= right; i++)
        System.out.print(a[top][i] + " ");
    top++;
    for (int i = top; i <= bottom; i++)
        System.out.print(a[i][right] + " ");
    right--;
    if (top <= bottom) {
        for (int i = right; i >= left; i--)
            System.out.print(a[bottom][i] + " ");
        bottom--;
    }
    if (left <= right) {
        for (int i = bottom; i >= top; i--)
            System.out.print(a[i][left] + " ");
        left++;
    }
}
}
}

```

OUTPUT:

Enter matrix elements:

1 2 3

4 5 6

7 8 9

Spiral Order:

1 2 3 6 9 8 7 4 5

SADDLE POINT:

```
import java.util.*;
```

```
class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int a[][] = new int[3][3];
```

```
        System.out.println("Enter matrix elements:");
```

```
        for(int i = 0; i < 3; i++) {
```

```

        for(int j = 0; j < 3; j++) {
            a[i][j] = sc.nextInt();
        }
    }
    boolean found = false;
    for(int i = 0; i < 3; i++) {
        int min = a[i][0];
        int col = 0;
        for(int j = 1; j < 3; j++) {
            if(a[i][j] < min) {
                min = a[i][j];
                col = j;
            }
        }
        boolean saddle = true;
        for(int k = 0; k < 3; k++) {
            if(a[k][col] > min) {
                saddle = false;
            }
        }
        if(saddle) {
            System.out.println("Saddle Point:" + min);
            found = true;
        }
    }
    if(!found) {
        System.out.println("No Saddle Point");
    }
    sc.close();
}
}

```

OUTPUT:

Enter matrix elements:

1 2 3

4 5 6

7 8 9

Saddle Point:7

PATTERNS:

PYRAMID:

```
import java.util.*;

class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");

        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {

            for (int j = 1; j <= n - i; j++) {

                System.out.print(" ");

            }

            for (int k = 1; k <= (2 * i - 1); k++) {

                System.out.print("*");

            }

            System.out.println();

        }

    }

}
```

OUTPUT:

Enter n: 6

```
      *

    ***

  *****

*****

*****

*****
```

HOLLOW PYRAMID:

```
import java.util.*;

class Main {
```

```

public static void main(String[] args) {
    Scanner sc=new Scanner(System.in);
    System.out.println("Enter n: ");
    int n=sc.nextInt();
    for(int i = 1; i <= n; i++) {
        for(int j = 1; j <= n - i; j++)
            System.out.print(" ");
        for(int k = 1; k <= (2*i-1); k++)
            if(k==1||k==(2*i-1)||i==n){
                System.out.print("*");
            }
        else{
            System.out.print(" ");
        }

        System.out.println();
    }
}

```

OUTPUT:

Enter n:

6

```

      *
     * *
    *   *
   *     *
  *       *
 *         *
*****

```